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SWITCHING CONVERTER SYSTEM

Abstract

A switching converter system comprising a first converter phase and a second converter phase, the switching converter system being operable in: a power conversion mode in which the first and second converter phases are operative to generate a multi-phase output voltage at an output of the switching converter circuit; and a calibration mode in which the first converter phase is operative as a reference supply for the second converter phase to cause a flow of current between the first and second converter phases to enable detection of a characteristic of each of the first and second converter phases.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure relates to calibration of power converter circuitry.

BACKGROUND

[0002] Power converters such as inductive boost converters typically include circuitry for sensing or monitoring a current through an energy storing circuit element (e.g. an inductor) of the power converter in one or more operational phases of the power converter.

[0003] One approach to sensing or monitoring such a current is to measure a voltage across an in-line current sense resistor (i.e. a resistor that is connected in series with the circuit element of interest). However, the use of such a current sense resistor can result in reduced efficiency, due to the power dissipated as heat by the current sense resistor.

SUMMARY

[0004] According to a first aspect, the invention provides a switching converter system comprising a first converter phase and a second converter phase, the switching converter system being operable in: a power conversion mode in which the first and second converter phases are operative to generate a multi-phase output voltage at an output of the switching converter circuit; and a calibration mode in which the first converter phase is operative as a reference supply for the second converter phase to cause a flow of current between the first and second converter phases to enable detection of a characteristic of each of the first and second converter phases.

[0005] The switching converter system may further comprise calibration control circuitry. In operation of the switching converter system in the calibration mode, the calibration control circuitry may be operative to: monitor current values of the first and second converter phases while applying a predefined adjustment to a parameter of the first converter phase; and detect the characteristic of each of the first and second converter phases based on the monitored current values and the predefined adjustment.

[0006] In operation of the switching converter system in the calibration mode, the calibration control circuitry may be further operative to: compare the detected characteristics of the first and second converter phases to respective predefined characteristics for the first and second converter phases; and apply a correction to the detected characteristic of the first and/or second converter phase if the detected characteristic does not correspond to the respective predefined characteristic.

[0007] In operation of the switching converter system in the calibration mode, an offset may be applied to a duty cycle of each of the first and second switching converter phases.

[0008] The offset may be applied to the duty cycle of each of the first and second switching converter phases as a result of the application of the predefined adjustment to the parameter of the first converter phase.

[0009] The current values of the first and second converter phases may comprise target current values of the first and second converter phases.

[0010] Applying the predefined adjustment to the parameter of the first converter phase may comprise applying a predefined gain adjustment to the first converter phase.

[0011] The first converter phase may comprise a first current sense path. The second converter phase may comprise a second current sense path. Applying a predefined adjustment to a parameter of the first converter phase may comprise applying a predefined gain adjustment to the first current sense path.

[0012] The first converter phase may comprise a first current sense path. The second converter phase may comprise a second current sense path. Applying a correction to the detected characteristic of the first and/or second converter phase if the detected characteristic does not correspond to the respective predefined characteristic may comprise applying a correction gain to

the first and/or second current sense path.

[0013] The first current sense path may comprise a first replica current sense path and the second current sense path may comprise a second replica current sense path.

[0014] The first replica current sense path may comprise: a replica device coupled to a switch of the first converter phase and configured to output a replica current indicative of a current through the switch; an amplifier for converting the replica current into a voltage; and a controllable gain stage for applying the correction gain;

[0015] The second replica current sense path may comprise: a replica device coupled to a switch of the second converter phase and configured to output a replica current indicative of a current through the switch; an amplifier for converting the replica current into a voltage; and a controllable gain stage for applying the correction gain.

[0016] The first converter phase may comprise a first control loop operative to generate the target current value of the first converter phase. The second converter phase may comprise a second control loop operative to generate the target current value of the second converter phase.

[0017] The first and second control loops may each comprise: a subtractor for generating a difference signal indicative of a difference between a sensed current through a switch of the respective first or second converter phase and a target current for the respective converter phase; a proportional-integral (PI) controller for generating a proportional-integral (PI) control signal phase on the difference signal; a multiplier for applying a signal indicative of a target output voltage for the respective converter phase to the PI control signal to modify the PI control signal based on the target output voltage.

[0018] The first and second converter phases may each comprise: a pulse-width modulator configured to generate a pulse-width modulated (PWM) control signal based on the modified PI control signal; and pre-driver circuitry configured to generate switch control signals for the switch of the respective first or second converter phase based on the PWM control signal.

[0019] The switching converter system may further comprise output voltage monitor circuitry configured for supplying a signal indicative of an output voltage of the first converter phase to the calibration control circuitry.

[0020] The calibration control circuitry may be operative to determine when an output voltage of the first converter phase has settled based on the signal indicative of the output voltage of the first converter phase, and to apply the predefined adjustment in response to a determination that the output voltage of the first converter phase has settled.

[0021] The first and second converter phases may be implemented in an integrated circuit.

[0022] The first converter phase may be implemented in a first integrated circuit and the second converter phase may be implemented in a second integrated circuit.

[0023] The first and second converter phases may each comprise boost converter circuitry, buck converter circuitry or buck-boost converter circuitry.

[0024] According to a second aspect, the invention provides a system comprising: first switching power converter circuitry having a first current sense path; second switching power converter circuitry having a second current sense path; and calibration control circuitry, wherein the system is operable in a calibration mode in which the first switching power converter circuitry is coupled to the second switching power converter circuitry to supply a current to the second switching power converter circuitry, wherein in operation of the system in the calibration mode, the first power converter is operated in a voltage regulation mode to output a target output voltage, and the calibration control circuitry is operative to: determine a first value of a first target current for the first switching power converter circuitry generated by a control loop of the first switching power converter circuitry; apply a predefined first value of a second target current for the second switching power converter circuitry to the second switching power converter circuitry; apply a predetermined gain adjustment to the first current sense path; determine a second value of the first target current generated by the control loop of the first switching power converter circuitry in

response to the gain adjustment; determine a second value of the second target current generated by a control loop of the second switching power converter circuitry in response to the gain adjustment; based on the first and second values of the first target current and the first and second values of the second target current, calculate a gain of the first current sense path and a gain of the second current sense path; and responsive to a determination that the calculated gain of the first or second current sense path is not equal to a predefined gain, apply a correction gain to the relevant one of the first or second current sense paths.

[0025] According to a third aspect the invention provides an integrated circuit comprising a switching converter system according to the first aspect or a system according to the second aspect.

[0026] According to a fourth aspect the invention provides a host device comprising a switching converter system according to the first aspect or a system according to the second aspect.

[0027] The host device may comprise a laptop, notebook, netbook or tablet computer, a gaming device, a games console, a controller for a games console, a virtual reality (VR) or augmented reality (AR) device, a mobile telephone, a portable audio player, a portable device, an accessory device for use with a laptop, notebook, netbook or tablet computer, a gaming device, a games console a VR or AR device, a mobile telephone, a portable audio player or other portable device.

[0028] Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0029] Embodiments of the invention will now be described, strictly by way of example only, with reference to the accompanying drawings, of which:

[0030] FIG. 1 is a simplified schematic representation of an inductive boost converter circuit that includes current monitoring circuitry that generates a replica current;

[0031] FIG. 2 is a schematic representation of a multi-phase switching power converter implementing a system according to the present disclosure; and

[0032] FIG. 3 is a schematic representation of a configuration of the multi-phase switching power converter of FIG. 2 for performing a field calibration operation.

DETAILED DESCRIPTION

[0033] An alternative approach to sensing or monitoring such a current through a circuit element is to generate a replica of a current through a switching element (e.g. a MOSFET device) that controls the current through the circuit element of interest, using one or more replica devices. The replica devices are typically smaller versions of the switching element that present a larger on-resistance than the switching element, such that the replica current that flows through the replica devices is a reduced-scale replica of the current through the switch. The replica current can be converted to a voltage using suitable current to voltage converter circuitry, and this voltage can be used to indicate the current through the circuit element of interest.

[0034] FIG. 1 is a simplified schematic representation of an inductive boost converter circuit that includes current monitoring circuitry that generates a replica current.

[0035] The inductive boost converter circuit, shown generally at **100** in FIG. 1, comprises an inductor **110**, a high-side switch **120**, a low-side switch **130** and an output capacitor **140**. The high-side switch **120** and the low-side switch **130** may be, for example, MOSFET devices.

[0036] The inductive boost converter circuit **100** further includes current monitoring circuitry **150**, comprising a replica device **152**, an amplifier **154** and an analog to digital converter (ADC) **156** which together comprise a current sensing path.

[0037] A first terminal of the inductor **110** is coupled to an input voltage node **112** at which, in use of the inductive boost converter circuit **100**, an input voltage V_{in} is received. The high-side switch **120** is coupled in series between a second terminal of the inductor and an output voltage node **114** at which, in use of the inductive boost converter **100**, an output voltage V_{Bst} is output. The low-side switch **130** is coupled in series between the second terminal of the inductor **110** and a ground (or other suitable reference voltage) supply rail **116**. The output capacitor **140** is coupled between the output voltage node **114** and the ground supply rail **116**.

[0038] In operation of the boost converter circuit **100**, during a first (charging) phase of operation, the low-side switch **130** is closed (switched on) and the high-side switch **120** is open (switched off) in response to suitable control signals output by controller circuitry (not shown in FIG. **1**), such that the inductor **110** is coupled between the input voltage node **112** and the ground supply rail **116**. In this first phase of operation, an increasing current flows through the inductor **110** and energy is stored in the inductor **110**.

[0039] In a second (discharging) phase of operation, the low side switch **130** is opened (switched off) and the high-side switch **120** is closed (switched on), such that the second terminal of the inductor **110** is coupled to the output voltage node **114**. Current can thus flow from the inductor **110** to the output capacitor **140** and to a load that may be coupled to the output voltage node **114**.

[0040] The replica device **152** of the current monitoring circuitry **150** is a physically smaller device of the same type (e.g. a MOSFET device) as the low-side switch **130**. When the low-side switch **130** is closed, a reduced-scale replica of the current that flows through the low-side switch **130** flows through the replica device **152**. This reduced-scale replica current is converted into an analog amplifier output voltage by the amplifier **154**, and this analog amplifier output voltage is converted into a digital signal indicative of the replica current (and thus of the current through the low-side switch **130**) by the ADC **156**.

[0041] Replica current monitoring circuitry of the kind described above with reference to FIG. **1** can be more efficient than current monitoring circuitry that uses an in-line current sense resistor, because there no power is dissipated as heat by a current sense resistor.

[0042] Replica current monitoring circuitry can suffer from gain errors, which may arise, for example, from the effects of different ageing of the replica device(s) (e.g. the replica device **152** in the example shown in FIG. **1**) and the circuit element of interest (e.g. the low-side switch **130**) or different electrical and/or mechanical stresses on the replica device(s) and the circuit element of interest. Such gain errors can lead to inaccuracy in the current measurements.

[0043] For example, in the inductive boost converter circuit **100** of FIG. **1**, the low-side switch **130** is physically larger than the replica device **152**. The size of the low-side switch **130** limits where it can be positioned in an integrated circuit (IC) implementation of the inductive boost converter circuit **100**, such that the low-side switch may have to be positioned in the vicinity of contact balls of the IC. As the replica device **152** is physically smaller than the low-side switch, there are fewer limitations as to the positioning of the replica device **152** in an IC, such that the replica device **152** can be positioned away from the contact balls. Mechanically-induced electrical stress (MIES) during packaging and soldering of the IC thus affects the low-side switch **130** differently than the replica device **152**. The MIES and subsequent trimming of the low-side switch **130** and the replica device **152** during production test can lead to part-to-part variations in the ratio of the impedance of the low-side switch **130** to that of the high-side switch. Such variations in the ratio can lead to reduced accuracy in current measurements.

[0044] Additionally, the replica device **152** ages differently than the low-side switch **130**. In operation of the inductive boost converter circuit **100** the low-side switch **130** is typically switched on and off by a high-frequency pulse width modulated (PWM) control signal, and a non-overlap period in which the low-side switch **130** is switched off occurs between every on-period of the low-side switch **130**. The effect of this is that the low-side switch **130** operates in its saturation region for very short periods of time, which causes electrical stress on the low-side switch **130** (arising, for

example, from effects such as Hot Carrier Injection (HCI) and Bias Temperature Instability (NBTI, PBOT), which increases the rate of ageing of the low-side switch **130**. In contrast, the replica device **152** is typically maintained in a switched-on state in operation of the inductive boost converter circuit **100**, to maximise the speed with which the current can be sensed, and thus the replica device **152** typically ages at a slower rate than the low-side switch **130**. These different rates of ageing can also cause variations in the ratio of the impedance of the low-side switch **130** to that of the high-side switch, again leading to reduced accuracy in current measurements.

[0045] Typically a calibration operation may be performed to calibrate out errors arising from MIES, component ageing and the like. In such a calibration operation a known reference signal, e.g. a reference current or voltage, is applied to a circuit and the circuit's response is measured. Parameters of circuit elements (e.g. a gain of an amplifier) are then adjusted to bring the circuit's response to the known reference signal into conformity with an expected response of the circuit to the reference signal.

[0046] However, for an inductive boost converter circuit of the kind described above with reference to FIG. 1, it may be difficult to perform such a calibration operation in the field, as it may not be possible to decouple the inductor **110** from the rest of the circuit. This makes it difficult to apply a known reference signal to the circuit.

[0047] The present disclosure provides a system and method for calibrating a first switching converter circuit in a multi-phase or multi-converter system using a second switching converter of the system to supply a reference signal, to compensate for an offset in the first switching converter.

[0048] FIG. 2 is a schematic representation of a multi-phase switching power converter implementing a system according to the present disclosure.

[0049] The multi-phase switching power converter shown generally at **200** in FIG. 2 includes a first switching power converter phase **210a** and a second switching power converter phase **210b**. The second switching power converter phase **210b** is identical in construction and operation to the first switching power converter phase **210a**. The following description of the first switching power converter phase **210a** thus also applies to the second switching power converter phase **210b**. For clarity, elements of the first switching power converter phase **210a** are denoted in FIG. 2 by a reference numeral with the suffix "a", and corresponding elements of the second switching power converter phase **210b** are denoted by the same reference numeral with the suffix "b".

[0050] In use of the multi-phase switching power converter **200**, the first and second switching power converter phases **210a**, **210b** each receive a supply voltage VBAT from a power supply **220**, which may be, for example, a battery.

[0051] The first switching power converter phase **210a** includes an inductor **222a** having a first terminal which is coupled, in use of the multi-phase switching power converter **200**, to the power supply **220** to receive the supply voltage VBAT.

[0052] A high-side switch **224a** (which may be, for example, a MOSFET device) is coupled between a second terminal of the inductor **222a** and an output node **230** of multi-phase switching power converter phase **200** (which is common to the first and second power converter phases **210a**, **210b**), such that when the high-side switch is closed (switched on), the second terminal of the inductor **222a** is coupled to the output node **230**. An output capacitor **232** is coupled between the output node **230** and a ground (or other reference voltage) supply.

[0053] A low-side switch **226a** (which may be, for example, a MOSFET device) is coupled between the second terminal of the inductor **222a** and the ground (other reference voltage) supply such that when the low-side switch **226a** is closed (switched on), the inductor **222a** is coupled between the power supply **220** and the ground supply to cause a current to flow through the inductor **222a**, storing energy in the inductor **222a**.

[0054] Operation of the high-side switch **224a** and the low-side switch **226a** is controlled by pre-driver circuitry **240a**, which outputs control signals to the high-side switch **224a** and the low-side switch **226a** cause them to open and close as appropriate to achieve a predefined target output

voltage (or output voltage range) at the output node 230. An operational cycle of the first switching power converter phase 210a includes a first, charging, stage in which suitable control signals are output by the pre-driver circuitry 240a to close (switch on) the low-side switch 226a and open (switch off) the high-side switch 224a to store energy in the inductor 222a and a second, discharging, stage in which are output by the pre-driver circuitry 240a close (switch on) the high-side switch 224a and open (switch off) the low-side switch 226a to allow current to flow from the inductor 222a to the output node 230 and to any load that may be connected to the output node 230.

[0055] The pre-driver circuitry 240a in turn receives pulse width modulated (PWM) control signals from digital pulse width modulator (DPWM) circuitry 242a, and generates the control signals for the high-side switch 224a and the low-side switch 226a based on the PWM control signals.

[0056] The first switching power converter phase 210a further includes a current sense path 250a comprising, in this example, one or more replica devices 252a, an amplifier 254a, an analog to digital converter (ADC) 256a, a low-pass filter 258a, a controllable digital gain stage 260a and a first digital multiplier 262a.

[0057] The one or more replica devices 252a are coupled to the high-side switch 224a and/or the low-side switch 226a and the amplifier 254a (e.g. as described above with reference to FIG. 1) so as to generate a reduced-scale replica of the current through the relevant switch 224a, 226a and the inductor 222a in operation of the first switching power converter phase 210a. The replica current is converted into an analog voltage by the amplifier 254a. The analog voltage is output by the amplifier 254a to the ADC 256a, which converts the analog voltage to a digital signal or code indicative of a sensed current through the relevant switch 224a, 226a and the inductor 222a during operation of the first switching power converter phase 210a.

[0058] As will be appreciated by those of ordinary skill in the art, if a replica device is coupled only to the low-side switch 226a, only the current through the low-side switch 226a and the inductor 222a in the first stage of an operational cycle of the first switching power converter phase 210a can be sensed or monitored, whereas if respective replica devices are coupled to both the high-side switch 224a and the low-side switch 226a, the current through the low-side switch 226a, the high-side switch 224a and the inductor 222a in both the first stage and the second stage of an operational cycle can be sensed or monitored.

[0059] The controllable digital gain stage 260a is operable to apply a controllable digital gain G1 to the filtered digital signal output by the digital low-pass filter 258a, via the first digital multiplier 262a. The controllable digital gain G1 is selected during a calibration operation to compensate for part-to-part variations in parameters of the circuit elements of the first power converter phase 210a, e.g. variations in the impedance of the high-side switch 224a, the low-side switch 226a or the replica device(s) 252a, or variations in a gain of the amplifier 254a or the digital low-pass filter 258a.

[0060] The first switching power converter phase 210a further includes a control loop including, in the illustrated example a digital subtractor 270a, a proportional-integral (PI) controller 272a and a second digital multiplier 274a. It will be appreciated by those of ordinary skill in the art that other types of controller could equally be used. For example, a proportional-integral-derivative (PID) controller, a type II controller or the like may be used in place of the PI controller 272a.

[0061] A first input of the digital subtractor 270a is coupled to an output of the first digital multiplier 262a to receive a gain-adjusted digital signal indicative of the sensed current through the inductor 222a during operation of the first switching power converter phase 210a. A second input of the digital subtractor 270a receives a digital signal ICMD indicative of a predefined target current through the inductor 222a. The digital subtractor 270a is configured to output a digital subtractor output signal indicative of a difference between the sensed current through the inductor 222a and the predefined target current through the inductor 222a to the PI controller 272a.

[0062] The PI controller 272a is operative to generate a PI control signal based on the digital

subtractor output signal, with the aim of minimising the difference between the sensed current through the inductor **222a** and the predefined target current through the inductor **222a**.

[0063] The PI control signal is output by the PI controller **272a** to a first input of the second digital multiplier **274a**. A second input of the second digital multiplier **274a** receives a digital signal $1/V_{BST}$ indicative of a value of a reciprocal of a predefined target output voltage for the first power converter phase **210a**. The second digital multiplier **274a** is thus operative to modify the PI control signal output by the PI controller **272a** by dividing it by the target output voltage. The second digital multiplier **274a** outputs this modified PI control signal to the digital PWM modulator **242a**, which generates the PWM control signals that are used by the pre-driver circuitry **240a** to generate control signals for the high-side and low-side switches **224a**, **226a** to regulate the current through the inductor **222a** and the output voltage of the first switching power converter phase **210a** in accordance with the predefined target inductor current and the predefined output voltage.

[0064] The multi-phase switching power converter **200** further includes calibration controller circuitry **280**. The calibration controller circuitry **280** may comprise a suitably configured microprocessor, microcontroller, state machine, application specific integrated circuit (ASIC), field programmable gate array (FPGA) or the like, or may comprise discrete circuitry or integrated circuitry configured to perform the operations described below.

[0065] A first input of the calibration controller circuitry **280** is coupled to an output of a supply voltage monitor ADC **282**. An input of the supply voltage monitor ADC **282** is coupled, in use of the multi-phase switching power converter **200**, to the power supply **220**. The supply voltage monitor ADC **282** is thus configured to generate and output a digital signal VPMON indicative of the supply voltage VBAT.

[0066] A second input of the calibration controller circuitry **280** is coupled to an output of an output voltage monitor ADC **284**. An input of the output voltage monitor ADC **284** is coupled, in use of the multi-phase switching power converter **200**, to the output node **230** of the first switching power converter phase **210a**. The output voltage monitor ADC **284** is thus configured to generate and output a digital signal VBSTMON indicative of the output voltage of the first power converter phase **210a**.

[0067] The calibration controller circuitry **280** has further inputs that are coupled, in use of the multi-phase switching power converter **200**, to outputs of the DPWM modulators **242a**, **242b** of the first and second switching power converter phases **210a**, **210b** respectively.

[0068] A first output of the calibration controller circuitry **280** is coupled to an input of the controllable digital gain stage **260a** of the first switching power converter phase **210a**. A second output of the calibration controller circuitry **280** is coupled to an input of the controllable digital gain stage **260b** of the second switching power converter phase **210b**.

[0069] The multi-phase switching power converter **200** further includes a calibration control path **290** comprising a calibration control path amplifier **292**, a calibration control path ADC **294** and a calibration control path PID controller **296**.

[0070] The multi-phase switching power converter **200** may be implemented as an integrated circuit (IC). In some examples the inductors **222a**, **222b** may be integrated in the IC (i.e. the inductors **222a**, **222b** may be on-chip devices). In other examples the inductors **222a**, **222b** may be separate from the IC (i.e. the inductors **222a**, **222b** may be off-chip devices).

[0071] Alternatively, the multi-phase switching power converter **200** may be implemented as a plurality of ICs, e.g. a first IC comprising the first switching power converter phase **210a** (with the inductor **222a** being either an on-chip device or an off-chip device) and a second IC comprising the second switching power converter phase **210b** (with the inductor **222b** being either an on-chip device or an off-chip device).

[0072] As a further alternative, the multi-phase switching power converter **200** may be implemented in discrete circuitry.

[0073] A production calibration operation may be performed in a production test environment after manufacture of the multi-phase switching power converter **200** (e.g. after fabrication of an IC implementing the multi-phase switching power converter **200**) to determine the gain of the current sense paths of each switching power converter phase **210a**, **210b**. In the event that the gain of one or both of the current sense paths **250a**, **250b**, as determined by the production calibration operation, does not correspond to a predefined expected gain for that current sense path, a correction gain can be applied by the controllable digital gain stage **260a**, **260b** of the relevant current sense path.

[0074] The production calibration operation is described below with reference to the first switching power converter phase **210a**. It will be appreciated that the same production calibration operation can be performed for the second switching power converter phase **210b**.

[0075] In the production calibration operation, a first known reference current (e.g. **1A**) is passed through the low-side switch **226a** and the output of the ADC **256a** is measured. The first known reference current is also passed through the high-side switch **224a** and the output of the ADC **256a** is measured. A second known reference current (e.g. **2A**) is passed through the low-side switch **226a** and the output of the ADC **256a** is again measured. The second known reference current is also passed through the high-side switch **224a** and the output of the ADC **256a** is measured.

[0076] From the measurements of the output of the ADC **256a** a low-side gain G_{LS1} and a high-side gain G_{HS1} of the current sense path of the first switching power converter phase **210a** are calculated (e.g. by the calibration controller circuitry **280** or by external processing circuitry of automated test equipment performing the production calibration operation). An overall gain $G1$ of the current sense path can also be calculated (e.g. by the calibration controller circuitry **280** or by external processing circuitry of the automated test equipment) using the relationship $G1 = G_{HS1} * (1 - D) + G_{LS1} * D$, where D is the duty cycle of a PWM control signal output by the DPWM modulator **242a**.

[0077] The low-side gain G_{LS2} , high-side gain G_{HS2} and overall gain $G2$ of the current sense path of the second switching power converter phase **210b** are also determined in a similar manner.

[0078] The measured outputs of the ADC **256a** for the first known reference current are stored, e.g. in a one-time programmable (OTP) memory element of the multi-phase switching power converter **200**, for subsequent use as a reference, e.g. to determine a change to one or more parameters of the first switching power converter phase **210a** that may arise due to ageing or degradation of one or more of the components of the first switching power converter phase **210a**.

[0079] The multi-phase switching power converter **200** is operable in a first, power conversion, mode, in which operation of the first and second switching power converter phases **210a**, **210b** is controlled by suitable control circuitry **205** (which may comprise, for example, discrete circuitry, integrated circuitry such as a microprocessor or microcontroller executing appropriate software commands, or an application specific integrated circuit or ASIC) such that an output voltage comprising a first output voltage component output by the first switching power converter phase **210a** and a second output voltage component output by the second switching power converter phase **210b**. Such an output voltage may be referred to as a multi-phase output voltage, as it includes components from multiple switching converter phases.

[0080] The multi-phase switching power converter **200** is also operable in a second, calibration, mode for performing a field calibration operation, as will now be described with reference to FIG. 3.

[0081] FIG. 3 shows a configuration of the multi-phase switching power converter **200** for use in a field calibration operation. In the example shown in FIG. 3, the first switching power converter phase **210a** acts as a current source and the second switching power converter phase **210b** acts as a current sink during the field calibration operation to cause a flow of current between the first and second switching power converter phases **210a**, **210b** to enable one or more characteristics of the first and second switching power converter phases **210a**, **210b** to be detected or determined. It will be appreciated, however, that the second switching power converter phase **210b** could equally act

as a current source and the first switching power converter phase **210a** could act as a current sink. [0082] The field calibration operation may be performed, for example, on power up or power-on reset of the multi-phase switching power converter **200**, and/or in dedicated maintenance or calibration mode in which the first and second phases **210a**, **210b** are calibrated.

[0083] As shown in FIG. 3, in use of the multi-phase switching power converter **200** for the field calibration operation, the output voltage of the first switching power converter phase **210a** is supplied to the high-side switch **224b** of the second switching power converter phase **210b**. Thus, in operation of the multi-phase switching power converter **200** in the second, calibration, mode, the first switching power converter phase **210a** acts as a reference supply for the second switching power converter phase **210b**.

[0084] An input of the output voltage monitor ADC **284** is coupled to the output node **230**, such that the output voltage monitor ADC **284** generates and outputs a digital signal VBSTMON indicative of the output voltage of the first switching power converter phase **210a** to the calibration controller circuitry **280**.

[0085] An input of the calibration control path amplifier **292** is also coupled to the output node **230**. An output of the calibration control path amplifier **292** is coupled to an input of the calibration control path ADC **294**. The calibration control path ADC **294** thus generates and outputs a digital signal indicative of the output voltage of the first power converter phase **210a**. An output of the calibration control path ADC **294** is coupled to an input of the calibration control path PID controller **296**. An output of the calibration control path PID controller **296** is coupled to a third input of the first digital multiplier **262b** of the second switching power converter phase **210b**. In some examples the output voltage monitor ADC **284** and the calibration control path ADC **294** may be implemented by the same ADC. For example, in some implementations in which the first and second switching power converter phases **210a**, **210b** are implemented in a single IC, the output voltage monitor ADC **284** and the calibration control path ADC **294** may be implemented by the same ADC.

[0086] The first switching power converter phase **210a** is operated (e.g. in response to suitable control signals output by the control circuitry **205**) in a voltage regulation mode to regulate the output voltage at the output node **230** at a target output voltage (e.g. 12V). In this mode of operation, the PWM signals output by the DWPM modulator **242a** and the target current digital signal ICMD1 can be adjusted by the control loop of the first switching power converter phase **210a** to meet the target output voltage.

[0087] The current $IL1$ through the inductor **222a** of the first switching power converter phase **210a** during the field calibration operation is defined by the relationship:

$IL1 = ICMD1 * G1$, (1) [0088] where $G1$ is the (unknown) gain of the current sense path of the first power converter phase **210a**.

[0089] A digital signal of value ICMD2 corresponding to a predefined known negative target current value (e.g. -1 A) through the inductor **222b** of the second switching power converter phase **210b** is input (e.g. by the control circuitry **205**) to the second input of the digital subtractor **270b** of the second power converter phase **210b**. The second switching power converter phase **210b** thus acts as a current sink during the field calibration operation.

[0090] The current through the inductor **222b** of the second switching power converter phase **210b** during the field calibration operation is defined by the relationship:

$IL2 = ICMD2 * G2$, (1) [0091] where $G2$ is the (unknown) gain of the current sense path of the second switching converter phase **210b**.

[0092] Once the output voltage at the output node **230** has settled (which may be determined, for example, by the calibration controller circuitry **280** comparing the digital signal VBSTMON to a digital signal indicative of the target output voltage), a steady state has been achieved by the first

switching power converter phase **210a** acting as a current source and the second switching power converter phase **210b** acting as a current sink. The calibration controller circuitry **280** then measures and stores a value ICMD1 of the target current signal ICMD for the first switching power converter phase **210a** and a value ICMD2 of the target current signal ICMD for the second switching power converter phase **210b**.

[0093] The calibration controller circuitry **280** then adjusts the gain G1 of the current sense path of the first switching power converter phase **210a** by a predetermined value ΔG (which may be positive or negative), such that the gain of the current sense path is $G1 + \Delta G$. This has the effect of changing the current through the inductor **222a**, and thus also changing the gain-adjusted digital signal indicative of the sensed current through the inductor **222a** output by the first digital multiplier **262a** to the digital subtractor **270a**. As a result, the digital subtractor output signal output by the digital subtractor **270a** changes. This change in the digital subtractor output signal causes the control loop of the first switching power converter phase **210a** to adjust the duty cycle of the PWM signals output by the DWPM modulator **242a** and the value of the target current signal ICMD1 to continue to meet the target output voltage.

[0094] Once the output voltage at the output node **230** has again settled at the target output voltage, the adjusted value ICMD1' of the target current signal ICMD1 for the first switching power converter phase **210a** is measured and stored by the calibration controller circuitry **280**.

[0095] As the value of the target current signal ICMD1 for the first switching power converter phase **210a** has been adjusted, a corresponding adjustment is required to the value of the target current signal ICMD2 for the second switching power converter phase **210b** to ensure that the second switching power converter phase **210b** is able to sink the current output by the first switching power converter phase **210a**, which also gives rise to a change in the duty cycle of the second switching power converter phase **210b**. The adjusted value ICMD2' of the value ICMD2 of the target current signal ICMD for the second switching power converter phase **210b** is measured and stored by the calibration controller circuitry **280**.

[0096] Thus, in operation of the switching power converter **200** in the second, calibration, mode, an adjustment or offset is applied to the duty cycles of the first and second switching converter phases **210a**, **210b**.

[0097] Defining the supply voltage provided by the power supply as VBAT and the output voltage of the first switching power converter phase **210a** as VBST, it can be shown that:

$$VBAT * IL1 * \eta_1 = VBST * i_{load1}; \text{ and} \quad (3)$$

$$VBAT * IL2 * \eta_2 = VBST * i_{load2}, \quad (4)$$

[0098] where η_1 is the efficiency of the first switching power converter phase **210a**, η_2 is the efficiency of the second switching power converter phase **210b**, i_{load1} is the load current of the first switching power converter phase **210a** and i_{load2} is the load current of the second switching power converter phase **210a**.

[0099] If $i_{load1} = i_{load2}$, then:

$$VBAT * IL1 * \eta_1 = VBAT * IL2 * \eta_2 \quad (5)$$

[0100] The efficiency η of a switching power converter can be expressed as:

$$\eta = (1 - D) * VBAT / VBST, \quad (6)$$

[0101] where D is the duty cycle of the switching power converter.

[0102] A ratio of the efficiency η_1 of the first switching power converter phase **210a** to the efficiency η_2 of the second switching power converter phase **210b** can be defined as:

$$\eta_1 / \eta_2 = (1 - D_1) / (1 - D_2), \quad (7) \text{ [0103] where } D_1 \text{ is the duty cycle of the first switching power}$$

converter phase **210a** [0104] and D2 is the duty cycle of the second switching power converter phase **210b**.

[0105] As noted above, $IL1=ICMD1*G1$ and $IL2=ICMD2*G2$.

[0106] From relationships (1), (2), (5) and (7) above it can be shown that:

$$G1/G2=(ICMD2/ICMD1)*(1-D2)/(1-D1) \quad (8)$$

Similarly:

$$(G1+\Delta G)/G2=(ICMD2'/ICMD1')*(1-D2')/(1-D1'), \quad (9)$$

[0107] where D1' and D2' are duty cycles of the first and second switching power converter phases, respectively, when the gain adjustment ΔG is applied.

[0108] Relationships (8) and (9) above are two linear equations for two unknowns (G1 and G2).

The calibration controller circuitry **280** can thus solve these equations to determine G1 and G2:

$$G1=K1*G2, \text{ and} \quad (10)$$

$$G1+\Delta G=K1'*G2. \quad (11)$$

[0109] From equations (10) and (11) above, it can be shown that:

$$G2=\Delta G/(K1'-K1), \text{ and} \quad (12)$$

$$G1=K1*\Delta G/(K1'-K1), \quad (13) \text{ [0110] where } K1=(ICMD2/ICMD1)*(1-D2)/(1-D1), \text{ and [0111] } K1'=(ICMD2'/ICMD1')*(1-D2')/(1-D1').$$

[0112] The calibration controller circuitry **280** calculates the gains G1 and G2 of the current sense paths of the first and second switching converter phases **210a**, **210b** respectively, based on equations (12) and (13) above.

[0113] In the absence of any gain errors in a current sense path (which, as explained above, may arise, for example, from the effects of ageing of the replica device(s) of the current sense paths or MIES), the gain of the current sense path should be equal to a predefined gain, which may be, for example, unity (i.e. 1 or 0 dB).

[0114] If the gain of a current sense path **250a**, **250b**, as calculated by the calibration controller circuitry **280** is not equal to the predefined gain, then there is a gain error in that current sense path.

[0115] In response to a determination, by the calibration controller circuitry **280**, that the calculated gain of a current sense path **250a**, **250b** does not correspond to the predefined gain for that current sense path **250a**, **250b**, the calibration controller circuitry **280** supplies a suitable control signal to the digital gain stage **260a**, **260b** of the affected current sense path **250a**, **250b** to cause that digital gain stage **260a**, **260b** to apply a correction gain to the relevant current sense path **250a**, **250b** to force the gain of the affected current sense path **250a**, **250b** to be equal to the predefined gain.

[0116] In this way the field calibration operation can be performed to correct any gain error in the current sense paths **250a**, **250b** without disconnecting the inductors **222a**, **222b** and without an external reference current.

[0117] The method described above can be extended to perform a field calibration operation for more than two switching power converter phases.

[0118] For example, first and second switching power converter phases may be configured to act as current sources, and third and fourth switching power converter phases may be configured to act as current sinks. In such an arrangement:

$$IL1+IL2=IL3=IL4,$$

where IL1 is the current through the inductor of the first switching power converter phase, IL2 is

the current through the inductor of the second switching power converter phase, IL3 is the current through the inductor of the third switching power converter phase and IL4 is the current through the inductor of the fourth switching power converter phase.

[0119] Predefined gain adjustments $\Delta G1$, $\Delta G2$ and $\Delta G3$ are applied to the current sense path of the first switching power converter phase, and the resulting target current values for the four switching power converter phases are measured by the calibration controller circuitry **280** to generate a system of four equations for four unknowns:

$$G1*ICMD1+G2*ICMD2=G3*ICMD3+G4*ICMD4$$

$$(G1+\Delta G1)ICMD1=G2*ICMD2'=G3*ICMD3'+G4*ICMD4'$$

$$(G1+\Delta G2)ICMD1=G2*ICMD2''=G3*ICMD3''+G4*ICMD4''$$

$$(G1+\Delta G3)ICMD1=G2*ICMD2'''=G3*ICMD3''' +G4*ICMD4''$$

(Note that in the system of equations above scale factors $(1-D2)/(1-D1)$ are omitted for simplicity. Those of ordinary skill in the art will appreciate that in a practical implementation such scale factors will be included in the system of equations).

[0120] The calibration controller circuitry **280** can then solve these equations for $G1$, $G2$, $G3$, and $G4$ to determine the gains of the current sense paths of the first, second, third and fourth switching power converter phases respectively, and can apply a correction gain to any current sense path where a gain error exists (i.e. any current sense path for which the calculated gain does not correspond to a predefined gain) as described above.

[0121] In examples where separate switching power converter phases are provided on separate ICs, primary current controller circuitry (which may be provided on one of the ICs, or may be separate from the ICs) may calculate the gains of the current sense paths as described above, and may transmit suitable control signals or instructions to the switching power converter phases on the separate ICs to apply any correction gain that may be required.

[0122] Although the description above relates to a multi-phase inductive boost converter, it will be appreciated that the system and method described above are equally applicable to other types of inductive power converter such as inductive buck converters and inductive buck-boost converters.

[0123] The example switching power converter **200** described above with reference to FIGS. 2 and 3 includes replica current sense paths for sensing the current through the inductors **226a**, **226b**.

However, it will be appreciated that alternative current sensing arrangements could equally be employed, e.g. using a current sense resistors coupled in series with the circuit element (e.g. inductor) of interest as described above, and suitable amplifier and/or signal processing circuitry for amplifying and/or processing the detected voltage across the current sense resistor to generate a signal indicative of the current through the circuit element of interest.

[0124] As will be appreciated by those of ordinary skill in the art, such alternative current sensing arrangements can also suffer from gain errors arising from, e.g. part to part variations in components, ageing and MIES.

[0125] The field calibration operation described above is equally applicable to such alternative current sensing arrangements to correct any gain error in current sense paths that include such alternative current sensing arrangements, again without disconnecting the inductors **222a**, **222b** and without an external reference current.

[0126] The circuitry described above with reference to the accompanying drawings may be incorporated in a host device such as a laptop, notebook, netbook or tablet computer, a gaming device such as a games console or a controller for a games console, a virtual reality (VR) or augmented reality (AR) device, a mobile telephone, a portable audio player or some other portable device, or may be incorporated in an accessory device for use with a laptop, notebook, netbook or

tablet computer, a gaming device, a VR or AR device, a mobile telephone, a portable audio player or other portable device.

[0127] The skilled person will recognise that some aspects of the above-described apparatus and methods may be embodied as processor control code, for example on a non-volatile carrier medium such as a disk, CD- or DVD-ROM, programmed memory such as read only memory (Firmware), or on a data carrier such as an optical or electrical signal carrier. For many applications embodiments of the invention will be implemented on a DSP (Digital Signal Processor), ASIC (Application Specific Integrated Circuit) or FPGA (Field Programmable Gate Array). Thus the code may comprise conventional program code or microcode or, for example code for setting up or controlling an ASIC or FPGA. The code may also comprise code for dynamically configuring re-configurable apparatus such as re-programmable logic gate arrays. Similarly the code may comprise code for a hardware description language such as Verilog™ or VHDL (Very high speed integrated circuit Hardware Description Language). As the skilled person will appreciate, the code may be distributed between a plurality of coupled components in communication with one another. Where appropriate, the embodiments may also be implemented using code running on a field-(re) programmable analogue array or similar device in order to configure analogue hardware.

[0128] Note that as used herein the term module shall be used to refer to a functional unit or block which may be implemented at least partly by dedicated hardware components such as custom defined circuitry and/or at least partly be implemented by one or more software processors or appropriate code running on a suitable general purpose processor or the like. A module may itself comprise other modules or functional units. A module may be provided by multiple components or sub-modules which need not be co-located and could be provided on different integrated circuits and/or running on different processors.

[0129] As used herein, when two or more elements are referred to as “coupled” to one another, such term indicates that such two or more elements are in electronic communication or mechanical communication, as applicable, whether connected indirectly or directly, with or without intervening elements.

[0130] This disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments herein that a person having ordinary skill in the art would comprehend. Similarly, where appropriate, the appended claims encompass all changes, substitutions, variations, alterations, and modifications to the example embodiments herein that a person having ordinary skill in the art would comprehend. Moreover, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, or component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Accordingly, modifications, additions, or omissions may be made to the systems, apparatuses, and methods described herein without departing from the scope of the disclosure. For example, the components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses disclosed herein may be performed by more, fewer, or other components and the methods described may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order. As used in this document, “each” refers to each member of a set or each member of a subset of a set.

[0131] Although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described above.

[0132] Unless otherwise specifically noted, articles depicted in the drawings are not necessarily drawn to scale.

[0133] All examples and conditional language recited herein are intended for pedagogical objects to aid the reader in understanding the disclosure and the concepts contributed by the inventor to furthering the art, and are construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the disclosure.

[0134] Although specific advantages have been enumerated above, various embodiments may include some, none, or all of the enumerated advantages. Additionally, other technical advantages may become readily apparent to one of ordinary skill in the art after review of the foregoing figures and description.

[0135] It should be noted that the above-mentioned embodiments illustrate rather than limit the invention, and that those skilled in the art will be able to design many alternative embodiments without departing from the scope of the appended claims. The word “comprising” does not exclude the presence of elements or steps other than those listed in a claim, “a” or “an” does not exclude a plurality, and a single feature or other unit may fulfil the functions of several units recited in the claims. Any reference numerals or labels in the claims shall not be construed so as to limit their scope.

Claims

1. A switching converter system comprising a first converter phase and a second converter phase, the switching converter system being operable in: a power conversion mode in which the first and second converter phases are operative to generate a multi-phase output voltage at an output of the switching converter circuit; and a calibration mode in which the first converter phase is operative as a reference supply for the second converter phase to cause a flow of current between the first and second converter phases to enable detection of a characteristic of each of the first and second converter phases.
2. A switching converter system according to claim 1, further comprising calibration control circuitry, wherein in operation of the switching converter system in the calibration mode, the calibration control circuitry is operative to: monitor current values of the first and second converter phases while applying a predefined adjustment to a parameter of the first converter phase; and detect the characteristic of each of the first and second converter phases based on the monitored current values and the predefined adjustment.
3. A switching converter system according to claim 2, wherein in operation of the switching converter system in the calibration mode, the calibration control circuitry is further operative to: compare the detected characteristics of the first and second converter phases to respective predefined characteristics for the first and second converter phases; and apply a correction to the detected characteristic of the first and/or second converter phase if the detected characteristic does not correspond to the respective predefined characteristic.
4. A switching converter system according to claim 2, wherein in operation of the switching converter system in the calibration mode, an offset is applied to a duty cycle of each of the first and second switching converter phases.
5. A switching converter system according to claim 4, wherein the offset is applied to the duty cycle of each of the first and second switching converter phases as a result of the application of the predefined adjustment to the parameter of the first converter phase.
6. A switching converter system according to claim 2, wherein the current values of the first and second converter phases comprise target current values of the first and second converter phases.
7. A switching converter system according to claim 2, wherein applying the predefined adjustment to the parameter of the first converter phase comprises applying a predefined gain adjustment to the first converter phase.

8. A switching converter system according to claim 2, wherein: the first converter phase comprises a first current sense path; the second converter phase comprises a second current sense path; and applying a predefined adjustment to a parameter of the first converter phase comprises applying a predefined gain adjustment to the first current sense path.
9. A switching converter system according to claim 3, wherein: the first converter phase comprises a first current sense path; the second converter phase comprises a second current sense path; and applying a correction to the detected characteristic of the first and/or second converter phase if the detected characteristic does not correspond to the respective predefined characteristic comprises applying a correction gain to the first and/or second current sense path.
10. A switching converter system according to claim 9, wherein the first current sense path comprises a first replica current sense path and the second current sense path comprises a second replica current sense path.
11. A switching converter system according to claim 9, wherein the first replica current sense path comprises: a replica device coupled to a switch of the first converter phase and configured to output a replica current indicative of a current through the switch; an amplifier for converting the replica current into a voltage; and a controllable gain stage for applying the correction gain; and wherein the second replica current sense path comprises: a replica device coupled to a switch of the second converter phase and configured to output a replica current indicative of a current through the switch; an amplifier for converting the replica current into a voltage; and a controllable gain stage for applying the correction gain.
12. A switching converter system according to claim 6, wherein: the first converter phase comprises a first control loop operative to generate the target current value of the first converter phase; and the second converter phase comprises a second control loop operative to generate the target current value of the second converter phase.
13. A switching converter system according to claim 12, wherein the first and second control loops each comprise: a subtractor for generating a difference signal indicative of a difference between a sensed current through a switch of the respective first or second converter phase and a target current for the respective converter phase; a proportional-integral (PI) controller for generating a proportional-integral (PI) control signal phase on the difference signal; a multiplier for applying a signal indicative of a target output voltage for the respective converter phase to the PI control signal to modify the PI control signal based on the target output voltage.
14. A switching converter system according to claim 13, wherein the first and second converter phases each comprise: a pulse-width modulator configured to generate a pulse-width modulated (PWM) control signal based on the modified PI control signal; and pre-driver circuitry configured to generate switch control signals for the switch of the respective first or second converter phase based on the PWM control signal.
15. A switching converter system according to claim 2, further comprising output voltage monitor circuitry configured for supplying a signal indicative of an output voltage of the first converter phase to the calibration control circuitry.
16. A switching converter system according to claim 15, wherein the calibration control circuitry is operative to determine when an output voltage of the first converter phase has settled based on the signal indicative of the output voltage of the first converter phase, and to apply the predefined adjustment in response to a determination that the output voltage of the first converter phase has settled.
17. A switching converter system according to claim 1, wherein the first and second converter phases are implemented in an integrated circuit.
18. A switching converter system according to claim 1, wherein the first converter phase is implemented in a first integrated circuit and the second converter phase is implemented in a second integrated circuit.
19. A switching converter system according to claim 1, wherein the first and second converter

phases each comprise boost converter circuitry, buck converter circuitry or buck-boost converter circuitry.

20. An integrated circuit comprising a switching converter system according to claim 1.

21. A host device comprising a switching converter system according to claim 1.

22. A host device according to claim 21, wherein the host device comprises a laptop, notebook, netbook or tablet computer, a gaming device, a games console, a controller for a games console, a virtual reality (VR) or augmented reality (AR) device, a mobile telephone, a portable audio player, a portable device, an accessory device for use with a laptop, notebook, netbook or tablet computer, a gaming device, a games console a VR or AR device, a mobile telephone, a portable audio player or other portable device.

23. A system comprising: first switching power converter circuitry having a first current sense path; second switching power converter circuitry having a second current sense path; and calibration control circuitry, wherein the system is operable in a calibration mode in which the first switching power converter circuitry is coupled to the second switching power converter circuitry to supply a current to the second switching power converter circuitry, wherein in operation of the system in the calibration mode, the first power converter is operated in a voltage regulation mode to output a target output voltage, and the calibration control circuitry is operative to: determine a first value of a first target current for the first switching power converter circuitry generated by a control loop of the first switching power converter circuitry; apply a predefined first value of a second target current for the second switching power converter circuitry to the second switching power converter circuitry; apply a predetermined gain adjustment to the first current sense path; determine a second value of the first target current generated by the control loop of the first switching power converter circuitry in response to the gain adjustment; determine a second value of the second target current generated by a control loop of the second switching power converter circuitry in response to the gain adjustment; based on the first and second values of the first target current and the first and second values of the second target current, calculate a gain of the first current sense path and a gain of the second current sense path; and responsive to a determination that the calculated gain of the first or second current sense path is not equal to a predefined gain, apply a correction gain to the relevant one of the first or second current sense paths.
